OPERATING SYSTEMS AND SYSTEMS PROGRAMMING

OVERVIEW

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Learning Objectives

General Organization of a Computer System and Interrupts

Components of Modern Multiprocessor Computer Systems

User Mode and Kernel Mode Transitions

Operating Systems in Various Computing Environments

Free and Open-Source Operating Systems

Resource Management: Process Management

An operating system is a resource manager. The system's CPU, memory space, file-storage space, and I/O devices are among the resources that the operating system must manage.

- A program can do nothing unless its instructions are executed by a CPU.
- A program in execution, as mentioned, is a process.
- A program such as a compiler is a process, and a word-processing program being run by an individual user on a PC is a process.
- A process needs certain resources—including CPU time, memory, files, and I/O devices—to accomplish its task.
- These resources are typically allocated to the process while it is running.
- In addition to the various physical and logical resources that a process obtains when it is created, various initialization data (input) may be passed along.

Resource Management: Process Management

- A program is a passive entity, like the contents of a file stored on disk, whereas a process is an active entity.
- A single-threaded process has one program counter specifying the next instruction to execute.
- The execution of such a process must be sequential.
- The CPU executes one instruction of the process after another, until the process completes.
- At any time, one instruction at most is executed on behalf of the process.
- Thus, although two processes may be associated with the same program, they are nevertheless considered two separate execution sequences.
- A multithreaded process has multiple program counters, each pointing to the next instruction to execute for a given thread.

Resource Management: Process Management

The operating system is responsible for the following activities in connection with process management:

- Creating and deleting both user and system processes,
- Scheduling processes and threads on the CPUs,
- Suspending and resuming processes,
- Providing mechanisms for process synchronization,
- Providing mechanisms for process communication

Resource Management: Memory Management

- Main memory is a large array of bytes, ranging in size from hundreds of thousands to billions.
- Each byte has its own address.
- Main memory is a repository of quickly accessible data shared by the CPU and I/O devices.
- The CPU reads instructions from main memory during the instruction-fetch cycle and both reads and writes data from main memory during the data-fetch cycle.
- To improve both the utilization of the CPU and the speed of the computer's response to its users, general-purpose computers must keep several programs in memory, creating a need for memory management.
- Many different memory management schemes are used.
- These schemes reflect various approaches, and the effectiveness of any given algorithm depends on the situation.

Resource Management: Memory Management

The operating system is responsible for the following activities in connection with memory management:

- Keeping track of which parts of memory are currently being used and which process is using them
- Allocating and deallocating memory space as needed
- Deciding which processes (or parts of processes) and data to move into and out of memory

Resource Management: File-System Management

- To make the computer system convenient for users, the operating system provides a uniform, logical view of information storage.
- The operating system abstracts from the physical properties of its storage devices to define a logical storage unit, the file.
- The operating system maps files onto physical media and accesses these files via the storage devices.
- The operating system implements the abstract concept of a file by managing mass storage media and the devices that control them.
- In addition, files are normally organized into directories to make them easier to use.

Resource Management: File-System Management

The operating system is responsible for the following activities in connection with file management:

- Creating and deleting files
- Creating and deleting directories to organize files
- Supporting primitives for manipulating files and directories
- Mapping files onto mass storage
- Mappingfilesonto mass storage
- Backing up files on stable (nonvolatile) storage media

Resource Management: Mass-Storage Management

- Most modern computer systems use HDDs and NVM devices as the principal on-line storage media for both programs and data.
- Most programs—including compilers, web browsers, word processors, and games—are stored on these devices until loaded into memory.
- The programs then use the devices as both the source and the destination of their processing.
- Hence, the proper management of secondary storage is of central importance to a computer system.
- The operating system is responsible for the following activities in connection with secondary storage management:
  - Mounting and unmounting
  - Free-space management
  - Storage allocation
  - Disk scheduling
  - Partitioning, Protection

Resource Management: Cache Management

- Caching is an important principle of computer systems. Here's how it works:
  - As it is used, it is copied into a faster storage system—the cache—on a temporary basis.
  - When we need a particular piece of information, we first check whether it is in the cache.
  - If it is, we use the information directly from the cache.
  - If it is not, we use the information from the source, putting a copy in the cache under the assumption that we will need it again soon.
- Because caches have limited size, cache management is an important design problem.
- Careful selection of the cache size and of a replacement policy can result in greatly increased performance
- In multiprocessor environments the value of a variable say "A" has to be updated on all the other local caches of other CPUs when its value is updated in a CPUs local cache (Cache coherency).

Resource Management: Cache Management

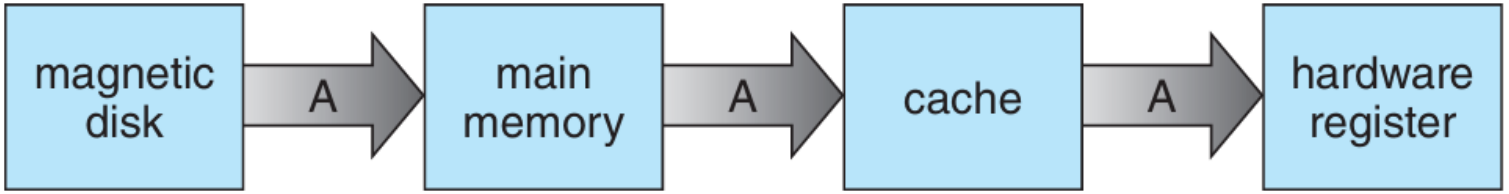

Figure 1: Migration of integer A from disk to register.

Level	1	2	3	4	5
Name	registers	cache	main memory	solid-state disk	magnetic disk
Typical size	< 1 KB	< 16MB	< 64GB	< 1 TB	< 10 TB
Implementation technology	custom memory with multiple ports CMOS	on-chip or off-chip CMOS SRAM	CMOS SRAM	flash memory	magnetic disk
Access time (ns)	0.25-0.5	0.5-25	80-250	25,000-50,000	5,000,000
Bandwidth (MB/sec)	20,000-100,000	5,000-10,000	1,000-5,000	500	20-150
Managed by	compiler	hardware	operating system	operating system	operating system
Backed by	cache	main memory	disk	disk	disk or tape

Figure 2: Characteristics of various types of storage.

Resource Management: I/O System Management

- One of the purposes of an operating system is to hide the peculiarities of specific hardware devices from the user.
- For example, in UNIX, the peculiarities of I/O devices are hidden from the bulk of the operating system itself by the I/O subsystem.
- The I/O subsystem consists of several components:
  - A memory-management component that includes buffering, caching, and spooling
  - A general device-driver interface
  - Drivers for specific hardware devices
- Only the device driver knows the peculiarities of the specific device to which it is assigned.

Virtualization

- Virtualization is a technology that allows us to abstract the hardware of a single computer into several different execution environments, thereby creating the illusion that each separate environment is running on its own private computer.
- These environments can be viewed as different individual operating systems (for example, Windows and UNIX) that may be running at the same time and may interact with each other.
- A user of a virtual machine can switch among the various operating systems in the same way a user can switch among the various processes running concurrently in a single operating system.
- Broadly speaking, virtualization software is one member of a class that also includes emulation.
- Emulation, which involves simulating computer hardware in software, is typically used when the source CPU type is different from the target CPU type.

Virtualization

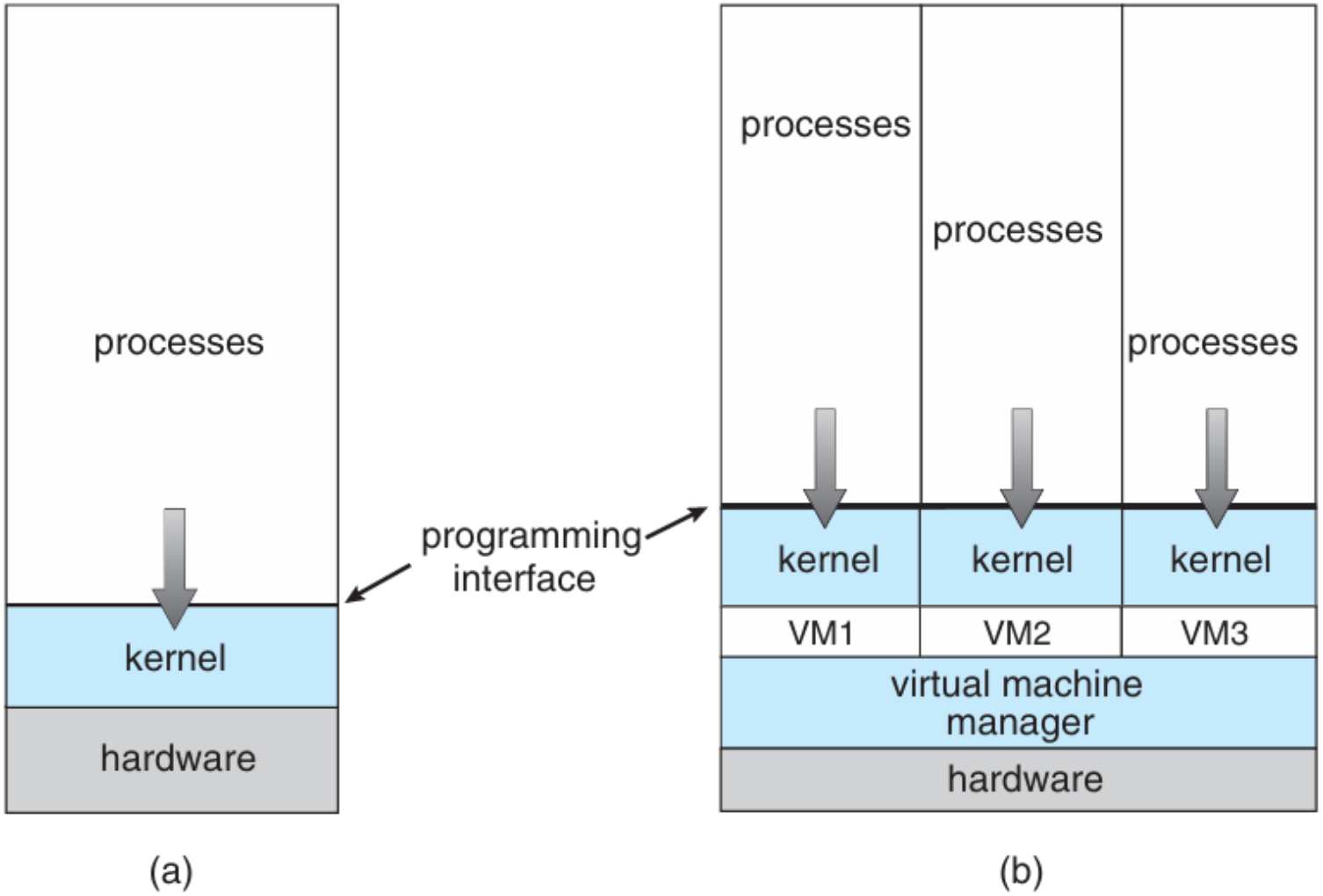

Figure 3: A computer running (a) a single operating system and (b) three virtual machines.

Distributed Systems

- A distributed system is a collection of physically separate, possibly heterogeneous computer systems that are networked to provide users with access to the various resources that the system maintains.
- Access to a shared resource increases computation speed, functionality, data availability, and reliability.

Computing Environments

Traditional Computing

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Thank you!

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